Activities 2016 / 2017

Nora Juhasz

March 29, 2017

Abstract

This document describes the university classes, GSSI classes, summer schools and seminars attended in the 2016 / 2017 academic year.

Contents

1 University of L'Aquila

1.1 Numerical Analysis 2 - Prof. Guglielmi

Euler method. Steepest descent gradient method. Conjugate directions method. Matlab.

1.2 High Performance Parallel Computing - Prof. Shcherbatyy, Prof. Pera

Basic Linux commands.

2 Gran Sasso Science Institute

2.1 Introduction to Finite Elements Method - Prof. Boffi

Finite differences, finite elements.

2.2 Decay property for hyperbolic systems with dissipation - Prof. Kawashima

The aim of the course is to study dissipative structure and decay property for hyperbolic systems with dissipation. Consider the following two equations.

symmetric diffusion	symmetric relaxation
$u_t + Au_x = Bu_{xx}$	$u_t + Au_x - Lu = 0$
B symmetric and nonnegative	L symmetric and nonnegative

2.3 Advanced Topics on Numerical Methods - Prof. Guglielmi

A-stability, B-stability, Runge-Kutta.

2.4 Introduction to the Incompressible Navier-Stokes equation and the mathematical theory of Turbulence - Prof. Marcati

3 Seminars

3.1 PDEs Day, 21st March 2017

3.1.1 Viscous vortex rings with axial symmetry - Prof. Gallay

3.1.2 A two phase model with cross and self attractive interactions - Prof. Novaga

3.1.3 Traveling waves for collective movements on a network - Prof. Corli

4 Summer schools

References

[Brezis, 2010] Brezis, H. (2010). *Functional analysis, Sobolev spaces and partial differential equations*. Springer Science & Business Media.

[Evans, 1990] Evans, L. C. (1990). *Weak convergence methods for nonlinear partial differential equations*. Number 74. American Mathematical Soc.

[Majda and Bertozzi, 2002] Majda, A. J. and Bertozzi, A. L. (2002). *Vorticity and incompressible flow*, volume 27. Cambridge University Press.